



Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	DDA6020
Course title (English)	Measure Theoretic Probability
Course title (Chinese)	基于测度理论的概率论
Units	3
Language of Instruction	English
Description (English)	Mesure theory: probability space, random variables, probability distributions, integration, Fubini's theorem, Lebesgue's dominated convergence theorem, conditional expectation; Various modes of convergence; weak and strong laws of large numbers, Borel-Cantelli lemmas; Characteristic functions and central limit theorems; Martingales and convergence theorems; Applications to Brownian motions and random walks.
Description (Chinese)	测度论：概率空间、随机变量、概率分布、积分、Fubini 定理、Lebesgue 界定收敛定理；条件期望；随机变量不同收敛方式；弱和强大数定理、Borel-Cantelli 引理；特征函数和中心极限定理；鞅及其收敛定理；对布朗运动和随机游走的应用。

2. Prerequisites / Co-requisites

A. Prerequisites

Calculus-based elementary probability and mathematical statistics, proof-based advanced calculus (or mathematical analysis) or real analysis.

B. Co-requisites

N/A

3. Learning Outcomes

Upon completion of the course, students will have acquired the ability to:



- Make rigorous probability arguments using the mathematical language of measure theoretic probability theory.
 - Understand essential concepts and results in measure theoretic probability and apply them to obtain theoretical results in application fields like operation research, machine learning, mathematical statistics and data science.
 - Understand various modes of convergence and prove convergence results.
 - Understand general conditional expectation and apply them in probability calculations.
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- Understand the proofs of the weak and strong laws of large numbers, and their applications.
 - Prove the central limit theorem using characteristic functions; apply the central limit theorem in application fields such as operation research, machine learning, mathematical statistics and data science.

4. Course syllabus

Measure theory Probability Spaces, Random Variables, Distributions, Integration, Fubini's theorem, Lebesgue's dominated convergence theorem, conditional expectation.

Law of large numbers Independence, various modes of convergence (convergence in probability, in distribution, and almost sure); Borel-Cantelli lemmas; weak and strong laws of large numbers.

Central limit theorems The De Moivre-Laplace Theorem, weak convergence; characteristic functions and central limit theorems; Doob's Inequality, convergence in L_p .

Application to Brownian motion and random walk. Brownian motion: definition and construction, Markov Property, Blumenthal's 0-1 law; Random walk: Donsker's theorem, CLT's for martingales.

5. Assessment Scheme

Component/ method	% weight
Coursework	30
Midterm Exam	30
Final Exam	40

6. Grade descriptor



General

Grade	Description
A	Demonstrates the ability to synthesize and apply the principles or subject matter learnt in the course, to novel situations and/or in novel ways, in a manner that would surpass the normal expectation at this level, and typical of standards that may be common at higher levels of study or research. Has the ability to express the synthesis of ideas or application in a clear and cogent manner.
A-	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to familiar and standard situations in a manner that is logical and comprehensive. Has the ability to express the knowledge or application with clarity.
B	Demonstrates the ability to state and partially apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is usually logically persuasive. Has the ability to express the knowledge or application in a satisfactory and unambiguous way.
C	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is not incorrect but is somewhat fragmented. Has the ability to express the separate pieces of knowledge in an unambiguous way.
D	Demonstrates the ability to state and sometimes apply the principles or subject matter learnt in the course to some simple and familiar situations in a manner that is broadly correct in its essentials Has the ability to state the knowledge or application in simple terms.
F	Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.

7. Feedback for evaluation

- Formal Course and Teaching Evaluation
- Informal feedbacks
- School retreat or program review

8. Reading

Required Reading



1. Probability: Theory and Examples (5th Edition)., 1108473687, Rick Durrett, Cambridge University Press, 5TH EDITION, 2019, England

9. Course components

Activity	Hours/week
Lecture	3
Self-study	3
Tutorial	1

10. Indicative teaching plan

Week	Content/ topic/ activity
1	Measures
2	Integration
3	Convergence theorems
4	Lp Spaces
5	Fubini's theorem
6	Foundations of Probability theory
7	Moments and generative functions
8	Independence
9	Borel-Cantelli Lemmas
10	Convergence of Random variables
11	Conditioning
12	Conditional expectations and Martingales
13	Stopping times
14	Decomposition of Martingales

11. Any other information

12. Version date

Version number	1
As of (date)	2025/08/13